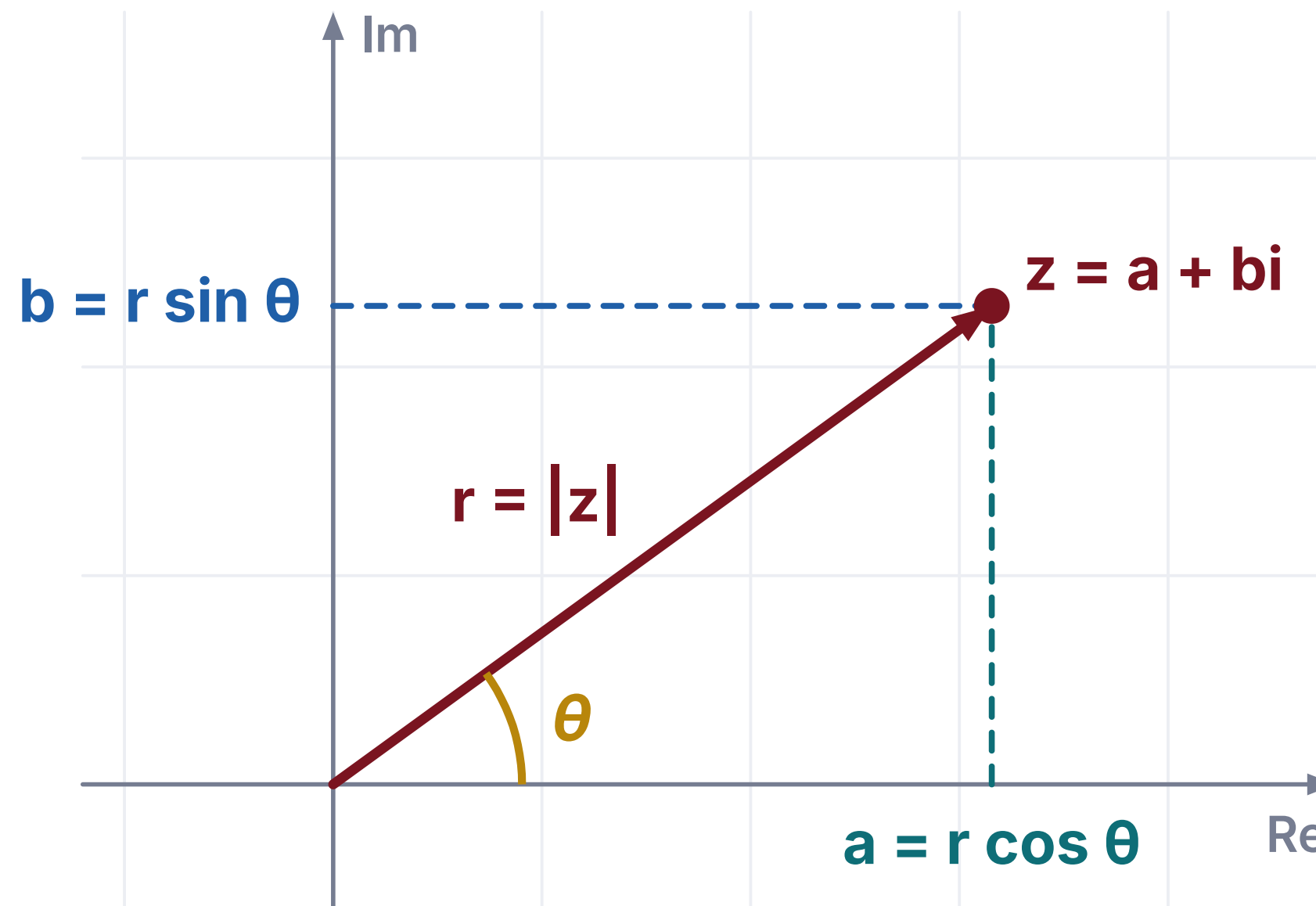


Euler's formula and the polar form

A second way to write a complex number: how far out, and which direction.

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Across and up, or out and round



From (r, θ) to (a, b) and back

$$a = r \cos \theta$$

$$b = r \sin \theta$$

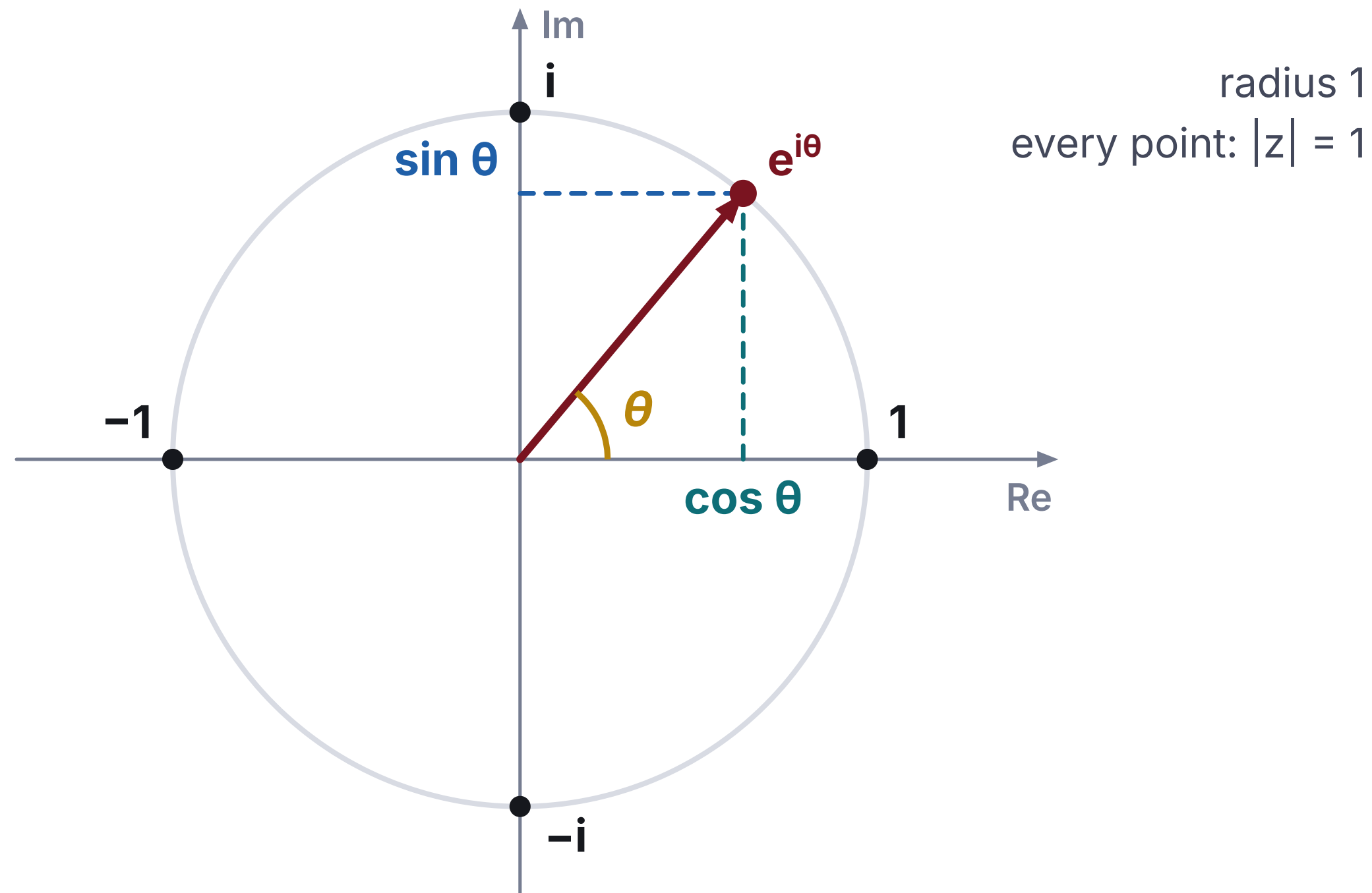
$$r = \sqrt{a^2 + b^2}$$

Angles are in **radians**, where $180^\circ = \pi$:

$$30^\circ = \pi/6 \quad 45^\circ = \pi/4$$

$$60^\circ = \pi/3 \quad 90^\circ = \pi/2$$

Every point at distance 1 is $\cos \theta + i \sin \theta$



$$e^{i\theta} = \cos \theta + i \sin \theta$$

Euler's formula. Take it as the meaning of $e^{i\theta}$; the proof needs calculus, which this course does not assume.

Why it deserves to be called an exponential

Exponents add when you multiply:

$$e^{i\alpha} e^{i\beta} = e^{i(\alpha+\beta)}$$

Check it with the school formulas for $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$.

$$\begin{aligned} & (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) \\ &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ &\quad + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ &= \cos(\alpha + \beta) + i \sin(\alpha + \beta) \end{aligned}$$

The polar form

$$z = r e^{i\theta}$$

The **modulus** $r = |z|$ is how far out.

The **phase** $\theta = \arg z$ is which direction, measured anticlockwise from the positive real axis.

Four points you should know by sight

$$e^{i0} = 1$$

$$e^{i\pi/2} = i$$

$$e^{i\pi} = -1$$

$$e^{i3\pi/2} = -i$$

Each one is $\cos \theta + i \sin \theta$ with a cos and sin you know.

For example, $e^{i\pi/2} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = 0 + i$.

A full turn, $e^{i2\pi}$, is back at 1.

$$e^{i\pi} = -1$$

Half a turn round the unit circle lands on -1 .

Write $z = 1 + i$ in polar form

The point $(1, 1)$ is in quadrant I, so the angle a calculator gives is the right one.

$$\begin{aligned}r &= \sqrt{1^2 + 1^2} = \sqrt{2} \\ \tan \theta &= 1/1 \Rightarrow \theta = \pi/4 \\ 1 + i &= \sqrt{2} e^{i\pi/4}\end{aligned}$$

Write $2e^{i\pi/3}$ in Cartesian form

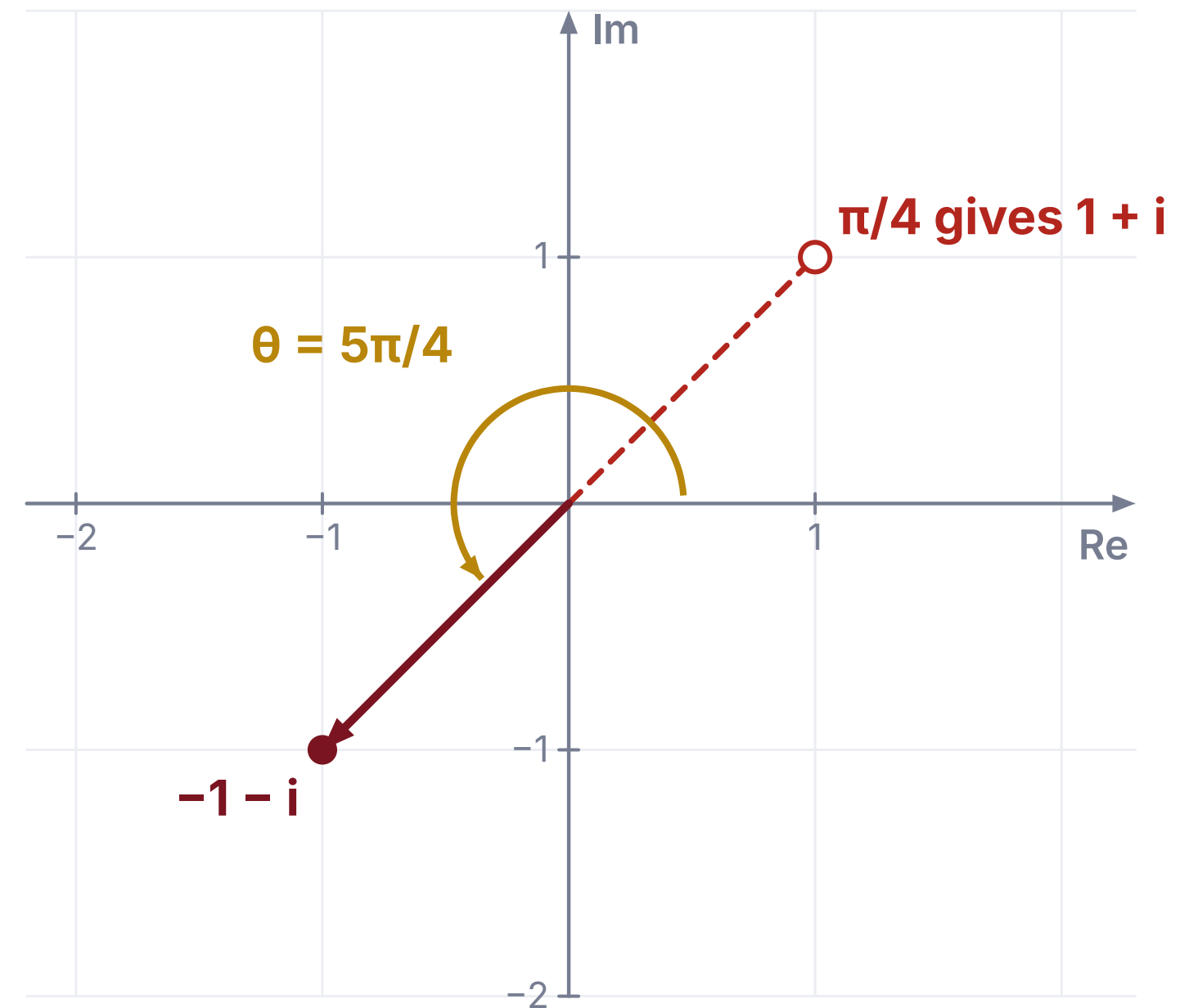
Use Euler's formula, then the values of $\cos 60^\circ$ and $\sin 60^\circ$.

$$\begin{aligned} 2e^{i\pi/3} &= 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ &= 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\ &= 1 + \sqrt{3}i \end{aligned}$$

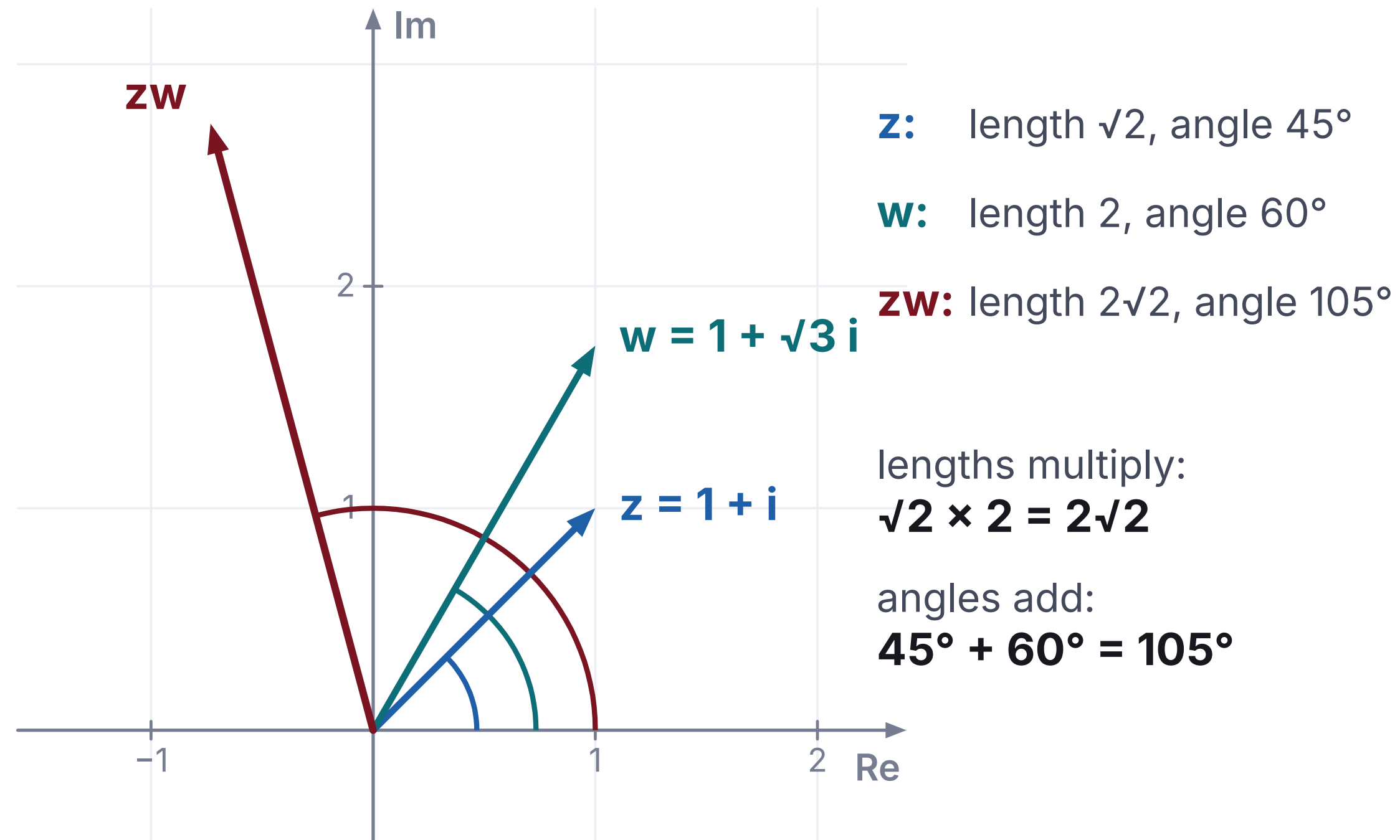
$\tan \theta = b/a$ cannot tell $1 + i$ from $-1 - i$

For $z = -1 - i$, $\tan \theta = 1$ and the calculator says $\pi/4$.
That is the direction of $1 + i$.

Always plot the point first. Here $\theta = 5\pi/4$, or equally $-3\pi/4$.



Lengths multiply. Angles add.



The same product, both ways

Polar:

$$\sqrt{2} e^{i\pi/4} \cdot 2 e^{i\pi/3} = 2\sqrt{2} e^{i7\pi/12}$$

since $\pi/4 + \pi/3 = 7\pi/12$, which is 105° .

Cartesian:

$$\begin{aligned} (1 + i)(1 + \sqrt{3}i) \\ &= 1 + \sqrt{3}i + i - \sqrt{3} \\ &= (1 - \sqrt{3}) + (1 + \sqrt{3})i \end{aligned}$$

Its modulus is $\sqrt{8} = 2\sqrt{2}$. They agree.

Compute $(1 + i)^8$

Expanding eight brackets is slow. In polar form a power is one line:

$$(re^{i\theta})^n = r^n e^{in\theta}$$

$$\begin{aligned}(1 + i)^8 &= \left(\sqrt{2} e^{i\pi/4}\right)^8 \\ &= (\sqrt{2})^8 e^{i2\pi} \\ &= 16 \cdot 1 = 16\end{aligned}$$

A phase factor turns, and never stretches

$$|e^{i\varphi}| = 1$$

So $e^{i\varphi}z$ is z turned by φ , same length. Multiplying by $i = e^{i\pi/2}$ is the quarter turn from Session 2.

In polar form the conjugate flips the angle:

$$z^* = re^{-i\theta}$$

$$z^*z = r^2$$

Add two amplitudes, then square

$$\begin{aligned}|z_1 + z_2|^2 &= (z_1 + z_2)^*(z_1 + z_2) \\&= |z_1|^2 + |z_2|^2 + z_1^* z_2 + z_2^* z_1 \\&= |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1^* z_2)\end{aligned}$$

The last term is the **cross term**. In polar form:

$$2 \operatorname{Re}(z_1^* z_2) = 2r_1 r_2 \cos(\theta_2 - \theta_1)$$

It depends only on the **difference** of the phases.

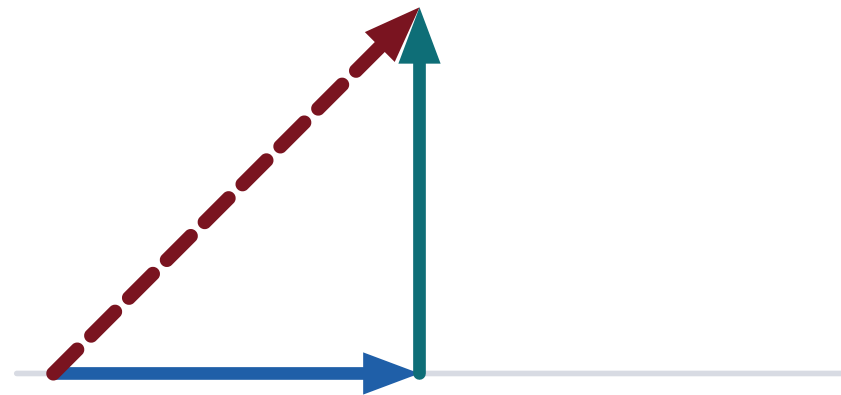
$z_1 = 1$ and $z_2 = e^{i\varphi}$: same sizes, different phase

$$\varphi = 0$$



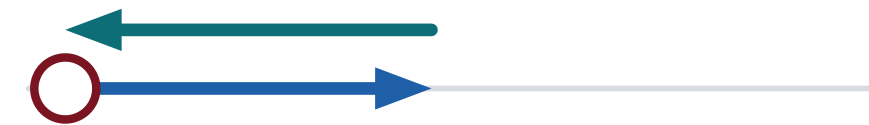
$$|z_1 + z_2|^2 = 4$$

$$\varphi = \pi/2$$



$$|z_1 + z_2|^2 = 2$$

$$\varphi = \pi$$



$$\text{sum} = 0$$

$$|z_1 + z_2|^2 = 0$$

One of these you can see, one you cannot

- **Global phase**

Multiply **both** by the same $e^{i\gamma}$:

$$|e^{i\gamma} z_1 + e^{i\gamma} z_2|^2 = |z_1 + z_2|^2$$

Nothing measurable changes.

- **Relative phase**

Change $\theta_2 - \theta_1$ and the cross term changes.

From 4 to 2 to 0 in the example.

This is the difference Unit 3 is built on.

Write $z = -\sqrt{3} + i$ in polar form

Then use the polar form to find z^6 .

Plot the point before you touch the calculator.

Four minutes. The calculator's first answer for θ is wrong here. Say why.

$$z = -\sqrt{3} + i$$

The modulus is $r = \sqrt{3 + 1} = 2$.

Then $\tan \theta = -1/\sqrt{3}$, and the calculator says $-\pi/6$: quadrant IV. The point is in quadrant II, so $\theta = \pi - \pi/6 = 5\pi/6$.

$$\begin{aligned} z &= 2 e^{i5\pi/6} \\ z^6 &= 2^6 e^{i5\pi} \\ &= 64 e^{i\pi} = -64 \end{aligned}$$

Modulus, phase and Euler in three calls

```
import numpy as np

z = -np.sqrt(3) + 1j
print(abs(z), np.angle(z))      # 2.0 2.6179938779914944 (5*pi/6)
print(np.angle(-1 - 1j))        # -2.356194490192345 (-3*pi/4, the same point as 5*pi/4)
print(np.exp(1j * np.pi))      # (-1+1.2246467991473532e-16j)
print(z ** 6)                   # (-63.99999999999997+1.4210854715202004e-14j)
```

Tiny leftovers such as `1e-16` are rounding error. `np.angle` always answers between $-\pi$ and π .

Before Session 4

Write $1 - i$, -2 and $3i$ in polar form.

Find $(1 - i)^6$ using the polar form.

For $z_1 = 1$ and $z_2 = i$, find $|z_1 + z_2|^2$ and the cross term.

Qubit Workbook, Part A (complex numbers, 10 marks): Sessions 1 to 3, due at the **end of week 4**.

Next: complex vectors.